Operational Roadmap for the Resolution of the Riemann Hypothesis: An AI-Driven Strategic Dossier

The Riemann Hypothesis (RH) has stood as the absolute zenith of unresolved conjectures in pure mathematics since its initial formulation by Bernhard Riemann in his seminal 1859 paper, *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse* (On the Number of Primes Less Than a Given Magnitude). Its resolution—whether by an affirmative deductive proof or the discovery of an explicit counterexample—carries cascading, structural implications for analytic number theory, the asymptotic distribution of prime numbers, quantum chaos theory, and the theoretical underpinnings of modern cryptographic systems. As the computational threshold crosses from brute-force heuristic search into the domain of autonomous, agentic artificial intelligence capable of formal verification and high-dimensional space exploration, a rigorously constructed operational roadmap becomes mathematically imperative.

This hyper-dense strategic research dossier establishes the foundational architecture for an advanced AI agent tasked with exploring the mathematical spaces surrounding the Riemann Hypothesis. To prevent the agent from expending computational cycles reinventing established mathematics, this report synthesizes a century and a half of human intuition. It meticulously translates complex-analytic topologies, functional analysis frameworks, noncommutative geometry, and arithmetic equivalences into discrete optimization tasks, formal verification paradigms, and generative search algorithms.

Phase 1: Foundations & Analytical Framework

To autonomously navigate the mathematical architecture surrounding the Riemann Hypothesis, an agentic system must possess an exhaustive, functionally operational understanding of the Riemann Zeta function, its topological constraints across the complex plane, and its absolute equivalences in both analytic error bounding and discrete arithmetic formulations.

The Riemann Zeta Function and the Mechanisms of Analytical Continuation

The Riemann Zeta function, denoted as $\zeta(s)$, is initially defined for a complex variable $s = \sigma + it$ (where $\sigma, t \in \mathbb{R}$) by the absolutely convergent Dirichlet series. Its profound connection to the discrete sequence of prime numbers is established through the Euler product formula, which holds strictly for $\mathrm{Re}(s) > 1$:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{n^{-s}}{1} = \prod_{p \text{ prime}} \left(1 - p^{-s}\right)^{-1}, \quad \text{for } \operatorname{Re}(s) > 1$$

This fundamental equivalence bridges the continuous geometry of complex analysis with the discrete arithmetic of prime factorization.[1] However, the Dirichlet series fails to converge for $\operatorname{Re}(s) \leq 1$. The mathematical architecture of RH relies entirely on the analytical continuation of $\zeta(s)$ to the entire complex plane. Riemann achieved this by utilizing the properties of the Gamma function $\Gamma(s)$ and contour integration, revealing that $\zeta(s)$ is a meromorphic function across the entire complex plane, containing a single, simple pole at $s = 1$ with a residue of 1.[1]

By introducing the completed zeta function, also known as Riemann's ξ-function, the underlying symmetry of the complex plane is established:

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s)$$

The function $\xi(s)$ is an entire function of order 1, devoid of any poles, which satisfies the highly symmetric functional equation:

$$\xi(s) = \xi(1-s)$$

This reflection formula dictates that the behavior of the zeta function in the right half-plane $\operatorname{Re}(s) > 1/2$ flawlessly dictates its behavior in the left half-plane $\operatorname{Re}(s) < 1/2$.[2] Because the Gamma function $\Gamma(s/2)$ possesses simple poles at negative even integers ($s = -2, -4, -6, \ldots$), $\zeta(s)$ must vanish at these precise locations to maintain the holomorphy of the entire $\xi(s)$ function. These specific roots are designated as the "trivial zeros" of the zeta function.

Topologies of the Critical Strip and the Critical Line

The trivial zeros offer no mystery regarding their location. The remaining zeros of $\zeta(s)$, known as the non-trivial zeros, are tightly constrained by the mathematical interaction between the functional equation and the Euler product. Because the Euler product is strictly non-zero for $\mathrm{Re}(s) > 1$, and the functional equation maps this non-zero region directly to the region $\mathrm{Re}(s) < 0$, it is a proven mathematical fact that all non-trivial zeros, denoted as $\rho = \beta + i\gamma$, must reside entirely within the open domain $0 < \mathrm{Re}(s) < 1$. This specific topological domain is defined as the **critical strip**.[1]

The Riemann Hypothesis posits a far more restrictive, singular constraint: all non-trivial zeros lie exactly on the one-dimensional subspace defined by $\mathrm{Re}(s) = 1/2$, a boundary known as the **critical line**.[2]

For an autonomous AI agent, the state space of the critical strip presents a rigorous, bounded topological geography. The absolute symmetry enforced by the functional equation ensures that if ρ is a non-trivial zero, then $1-\rho$, the complex conjugate $\overline{\rho}$, and $1-\overline{\rho}$ are also zeros.[2] Therefore, any non-trivial zero located strictly off the critical line (for instance, at a hypothetical $\beta = 0.6$) would immediately generate a symmetric quadruplet of zeros. This geometric arrangement would break the known density expectations of zeros on the critical line and instantly serve as an explicit counterexample to the hypothesis.

Equivalence with the Prime Number Theorem

The Riemann Hypothesis is intimately linked to the error term embedded within the Prime Number Theorem (PNT). The PNT establishes the asymptotic density of primes, asserting that the prime-counting function $\pi(x)$, which counts the number of primes less than or equal to x, behaves asymptotically as the logarithmic integral:

$$\pi(x) \sim \mathrm{Li}(x) = \int_2^x \frac{dt}{\ln t}$$

The profound connection to the zeros of the Riemann zeta function is encoded dynamically in von Mangoldt's explicit formula for the Chebyshev function $\psi(x) = \sum_{p^k \le x} \ln p$:

$$\psi(x) = x - \sum_{\rho} \frac{x^\rho}{\rho} - \ln(2\pi) - \frac{1}{2}\ln(1 - x^{-2})$$

The magnitude of the oscillatory term $\sum \frac{x^\rho}{\rho}$ is governed entirely by the real parts of the non-trivial zeros. The Riemann Hypothesis is mathematically equivalent to the statement that the error term in the Prime Number Theorem is bounded by the square root of x (up to logarithmic factors) [4]:

$$\pi(x) = \mathrm{Li}(x) + \mathcal{O}(\sqrt{x}\ln x)$$

For a computational AI agent, this translates the highly abstract complex-analytic problem into a concrete real-variable bounding problem: discovering any deviation in prime distribution that mathematically exceeds the $\mathcal{O}(\sqrt{x}\ln x)$ boundary constitutes an absolute falsification of RH.[6]

Arithmetic Equivalences: Robin, Lagarias, Guy, and Nicolas

To provide discrete, finite-state spaces suitable for search-based AI agents, the Riemann Hypothesis can be reframed entirely into purely elementary arithmetic inequalities. An agent searching for a counterexample can optimize search parameters against these rigorously established bounds.

Criterion Name	**Mathematical Formulation**	**Range of Validity for RH**	**Domain of Application & Metrics**
Robin's Criterion (1984)	$\sigma(n) <$	For all $n > 5040$	Utilizes the sum of divisors function $\sigma(n)$, where $\gamma \approx$ is the Euler-Mascheroni constant.[7]
Lagarias' Criterion (2002)	$\sum_{d	n} d \le H_n + \exp(H_n)\ln(H_n)$	For all integers $n \ge$
Nicolas' Criterion (1983)	$\frac{N_k}{\varphi(N_k)} > e^{\gamma} \ln(\ln N$	For all integers $k \ge$	Utilizes Euler's totient function $\varphi(n)$ evaluated strictly at primorials $N_k = \prod_{i=1}^{k} p_i$

			.[13]
Guy's Formulation	Bounds related to Aliquot sequences.	Unbounded sequences.	Explores the sum of proper divisors, questioning whether sequences like $s(n)$ can escape strict bounds linked to RH.[16]

The **Nicolas Criterion** is of paramount interest for high-dimensional computational AI searches. The inequality rigorously asserts that RH holds if and only if the Euler totient bound strictly exceeds the specified logarithmic threshold for all primorial numbers N_k.[13] Recent theoretical probes suggest that if a counterexample to RH exists, it will not be found among small integers; it must emerge as a highly composite number or an exceptionally large primorial, with modern estimates requiring computational searches where the k-th prime $p_k > 10^8$.[13] An AI agent equipped with advanced sieving algorithms and arbitrary-precision integer representations can target these specific arithmetic spaces, transforming a continuous complex analysis problem into an integer optimization anomaly search.

Similarly, Robin's criterion forces the agent to analyze the sum of divisors $\sigma(n)$ against the Euler-Mascheroni constant bounds.[7] If an AI system discovers a single integer $n > 5040$ where $\sigma(n) \geq e^{\gamma} n \ln(\ln n)$, the Riemann Hypothesis is instantly proven false.[10] Current computational efforts have verified Lagarias' and Robin's criteria up to extreme limits, but the unbounded nature of integers requires intelligent, targeted heuristic generation rather than pure brute force.[8]

Phase 2: The Shoulders of Giants (Historical Paradigms)

An autonomous mathematical agent cannot afford to traverse the infinite landscape of RH blindly. It must leverage the deep structural paradigms that have shaped modern analytic number theory. Four primary mathematical frameworks dominate the contemporary approach to the Riemann Hypothesis, each presenting distinct computational advantages, unresolved theoretical bottlenecks, and potential algorithmic entry points.

1. The Hilbert-Pólya Conjecture & Random Matrix Theory

In the early 20th century, the Hilbert-Pólya conjecture postulated a profound spectral interpretation for the non-trivial zeros of the zeta function. It suggested the existence of a self-adjoint quantum operator $\mathcal{H}$ (a Hamiltonian) whose eigenvalues λ correspond precisely to the imaginary parts of the non-trivial zeros, taking the form $\rho = 1/2 + i\lambda$. Because self-adjoint operators possess strictly real eigenvalues, the existence of such an operator would immediately, by definition, prove that all non-trivial zeros possess a real part exactly equal to $1/2$.[20]

This conjecture found astonishing validation in the 1970s through Random Matrix Theory (RMT). Hugh Montgomery formulated the pair correlation conjecture, demonstrating mathematically that the normalized spacing between consecutive zeros of $\zeta(s)$ on the critical line follows the exact same statistical distribution as the eigenvalue spacings of large random Hermitian matrices drawn from the Gaussian Unitary Ensemble (GUE).[1] This connection between prime number distributions and quantum physics was later verified computationally to staggering precision by Andrew Odlyzko, who calculated the spacing of millions of zeros near the 10^{20}-th zero, finding the GUE statistics to be a nearly perfect match.[22]

Structural Advantage: Random Matrix Theory provides a highly robust probabilistic model for zero distribution, allowing statistical algorithms to predict zero clustering, spacing, and long-range repulsion accurately.[20] **Unresolved Bottleneck:** The matrices in the GUE are purely statistical aggregates. No explicit, deterministic operator $\mathcal{H}$ associated with the integers or the prime numbers has been universally accepted or discovered. The theoretical gap between statistical correlation and the exact geometric construction of a self-adjoint operator remains the primary obstruction. An AI must attempt to synthesize this operator rather than merely confirming its statistical echoes.

2. Noncommutative Geometry (Alain Connes' Approach)

Fields Medalist Alain Connes radically advanced the Hilbert-Pólya vision by constructing an explicit spectral realization of the Riemann zeros using the advanced tools of Noncommutative Geometry (NCG).[23] Connes constructed a quantum mechanical space—specifically the adele class space of the rational numbers $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$—where the non-trivial zeros of $\zeta(s)$ appear definitively as an absorption spectrum associated with a specific trace formula.[25]

In this paradigm, the Weil explicit formula from algebraic geometry is interpreted dynamically as a Lefschetz trace formula on this noncommutative space. The proof of the Riemann Hypothesis is thus reduced geometrically to establishing the positivity of the Weil distribution (Weil positivity) across a specific Hilbert space.[25]

Structural Advantage: Connes successfully mapped the global Riemann-Weil explicit formula into a rigorous geometric setting, creating what is known as the "Scaling Hamiltonian".[27] This shifts the problem into the realm of operator algebras and functional limits. **Unresolved Bottleneck:** The original 1999 formulation suffered from a severe structural difficulty: the trace formula interpretation struggled with an arbitrary cutoff parameter δ and the potential presence of non-critical zeros appearing as false resonances.[25] To bypass this bottleneck, Connes and Caterina Consani refined the approach to focus strictly on the semi-local trace formula. They identified the "Sonin space" (a specialized subspace of test functions whose Fourier transforms vanish at $\pm i/2$) as the core domain necessary to define the negative part of the spectrum.[27] Proving Weil positivity over the semilocal Sonin space remains the critical unscaled cliff. For an AI agent, the generation of highly specific test functions $g(x) \in C_c^\infty(\mathbb{R}_+^\times)$ within the Sonin space that violate this positivity acts as a targeted counterexample generation protocol.[27]

3. The Beurling-Nyman Criterion (Functional Analysis in L^2)

The Beurling-Nyman formulation entirely shifts the battlefield of RH from complex analysis to the functional analysis of Hilbert spaces. Originally proven by Nyman in his 1950 thesis and subsequently generalized by Beurling in 1955, the criterion states that the Riemann Hypothesis is mathematically equivalent to the assertion that the characteristic indicator function of the interval $(0, 1]$, denoted mathematically as $\chi_{(0,1)}$, lies within the closed linear span in the $L^2(0, 1)$ space of a specific subspace of fractional part functions.[34]
Luis Báez-Duarte strengthened this theorem dramatically in 2003, showing that one need only consider simple integer dilations. Specifically, RH is equivalent to the statement that the distance d_N between the indicator function $\chi_{(0,1)}$ and the linear span of functions $f_k(t) = \{t/k\}$ (where $\{x\}$ denotes the fractional part of x) strictly approaches zero as $N \to \infty$.[35]

The squared distance is formalized as an infimum over all Dirichlet polynomials $A_N(s)$ of length N:

$$d_N^2 = \inf_{A_N} \frac{1}{2\pi} \int_{-\infty}^{\infty} \left|1 - \zeta\left(\frac{1}{2} + it\right) A_N\left(\frac{1}{2} + it\right)\right|^2 \frac{dt}{1/4 + t^2}$$

Structural Advantage: This paradigm beautifully frames RH as a strictly real-variable, least-squares approximation optimization problem inside a well-understood Hilbert space.[39] AI models are intrinsically designed for high-dimensional least-squares minimization. **Unresolved Bottleneck:** The Vasyunin sums, which directly dictate the optimal coefficients a_k for the

Dirichlet polynomials, are exceptionally non-linear and difficult to optimize at large N.[37] Current theoretical bounds suggest the distance drops at an agonizingly slow rate. For instance, assuming RH is true, bounds show $d_N^2 \leq (\log \log N)^{5/2+o(1)} (\log N)^{-1/2}$.[38]

AI agents excel at L^2 parameter inference; the calculation of optimal Vasyunin sums via machine learning or evolutionary algorithms presents a highly viable entry point for generating a definitive approximation sequence.

4. Universality and the Dirichlet L-functions

The Riemann Zeta function does not exist in a mathematical vacuum; it is the simplest archetype of the broader Selberg class of L-functions, which importantly includes Dirichlet L-functions associated with arithmetic characters.[41] The Generalized Riemann Hypothesis (GRH) asserts that all L-functions within this expansive class share the same fundamental property: their non-trivial zeros lie strictly on their respective critical lines.[36]

A deeply counterintuitive and complex property of these functions is encapsulated in Sergei Voronin's Universality Theorem (1975). Voronin proved that the Riemann zeta function is "universal" in the right half of the critical strip ($1/2 < \mathrm{Re}(s) < 1$). Specifically, any non-vanishing, continuous holomorphic function defined on a compact subset within this specific domain can be approximated arbitrarily well by suitable purely imaginary vertical translations of the zeta function, $\zeta(s + i\tau)$.[42]

Structural Advantage: Universality implies that the local behavior of $\zeta(s)$ is infinitely chaotic and rich, containing the geometry of all possible analytic functions within its vertical shifts.[43] This offers a massive parametric space for an AI to generate function approximations.

Unresolved Bottleneck: Because $\zeta(s)$ can theoretically mimic functions with zeros that are arbitrarily close to the critical line, proving a strict lower bound that pushes all zeros identically onto the line requires separating the chaotic, universal behavior on the right from the rigid, algebraic structure of the critical line itself. Furthermore, Davenport-Heilbronn L-functions—which possess functional equations but critically lack Euler products—are explicitly known to have zeros off the critical line, breaking RH.[1] Thus, any AI proof strategy must explicitly and deeply encode the Euler product to logically separate $\zeta(s)$ from its non-RH-compliant cousins.

Phase 3: AI-Driven Metaphorical Optimization ("Antigravity Landscapes")

For an autonomous agent to process the immense weight of these historical paradigms, abstract algebraic geometry and complex topology must be computationally translated into quantifiable, programmable frameworks. This necessitates mapping the search for zeros—or the verification of formal proofs—into explicit optimization domains suitable for Deep Reinforcement Learning (DRL) and mechanized formal verification engines.

Simulating Repulsive Forces ("Antigravity") via Coulomb Gas Models

If the critical line $\mathrm{Re}(s) = 1/2$ acts mathematically as an inescapable "gravitational well" capturing all non-trivial zeros, we must mathematically parameterize the topology of this well. Taking direct inspiration from the GUE matrix statistics, the zeros can be modeled as a one-dimensional Coulomb gas of point charges confined to the real line (acting as the critical line after a 90-degree phase shift in the complex plane). Dyson's Brownian motion model dictates that these charges undergo stochastic dynamics while exerting a mutual, logarithmic electrostatic repulsion upon one another.[20]

An AI agent can exploit this by utilizing the Hardy Z-function, defined mathematically as $Z(t) = \zeta(1/2 + it)e^{-i\theta(t)}$, where $\theta(t)$ is the Riemann-Siegel theta function.[47] The beauty of the Hardy function is that $Z(t)$ is entirely real for real values of t, and its sign changes correspond precisely and exclusively to the zeros of $\zeta(s)$ on the critical line. The exact locations where $\theta(g_n) = \pi n$ are known as Gram points, g_n. "Gram's Law" historically posited that $(-1)^n Z(g_n) > 0$. If this law held universally for all n, RH would be trivially true. However, the law explicitly fails at infinite discrete locations known mathematically as "bad Gram points".[47]

To test computationally if zeros can be ejected off the critical line (creating an algorithmic "antigravity" scenario), the agent should evaluate the discriminant function $\Delta_n(a)$ precisely at these bad Gram points. Recent theoretical advancements demonstrate that the second-order approximation of this highly non-linear discriminant functions exactly as the repulsion term in a Dyson Coulomb gas.[47]

The AI Translation: In a Deep Reinforcement Learning architecture, the state space consists of sequences of bad Gram points g_n. The agent's action space allows it to dynamically perturb the continuous variables $a \in \mathbb{R}^{N(n)}$ within the discriminant function Δ_n. The internal reward function is structurally mapped directly to the Coulomb potential energy E. An "antigravity" reward is achieved if the agent successfully identifies a configuration where the local zero repulsion mathematically overcomes the global structural confinement. Finding a parameter set where the corrected Gram's law $(-1)^n \Delta_n(1) > 0$ is explicitly violated [47] would directly correlate with a zero existing at $\mathrm{Re}(s) \neq 1/2$. The agent is thus playing a game of electrostatic manipulation against the fabric of the zeta function.[1]

Formal Verification Architecture: Lean 4 and Isabelle/HOL

While Deep Reinforcement Learning provides an empirical mechanism for locating counterexamples or discovering heuristic bounds, an autonomous agent tasked with proving the Riemann Hypothesis definitively must construct deductively unassailable, logically airtight chains. This absolute requirement necessitates the use of interactive theorem provers (ITPs) such as Lean 4 or Isabelle/HOL.

In 2025, David Loeffler and Michael Stoll achieved a monumental milestone by formalizing the foundational theory of the Riemann zeta function, Dirichlet L-functions, and the precise formal statement of the Riemann Hypothesis within Lean 4's expansive Mathlib library.[41] The formal architecture they built includes:

1. riemannZeta: A rigorous formal definition of the function across the complex plane.[50]
2. riemannZeta_eulerProduct: The formal proof of the product formula, securing the prime connection.[50]
3. riemannZeta_one_sub: The formalized functional equation relating $\zeta(s)$ to $\zeta(1-s)$.[50]

The ultimate target state, RiemannHypothesis, rigorously asserts in Lean that all zeros of $\zeta(s)$, excluding the trivial zeros at negative even integers, possess a real part exactly equal to $1/2$.[50] Furthermore, Gomes and Kontorovich previously formalized a logically equivalent statement using the Dirichlet η function [50], while Manuel Eberl formalized substantial tracts of analytic number theory within the Isabelle theorem prover.[50]

The AI Translation: An agentic AI system, utilizing neural-symbolic frameworks such as Lean Dojo [52], interfaces directly with the Lean 4 environment as a formalized game.

1. **State Space:** The current tactic state, comprised of a list of active hypotheses and the mathematical target goal.
2. **Action Space:** The systematic application of mathematically valid Lean tactics (e.g., rw, apply, exact, induction, simp).
3. **Reward:** The successful closing of a logical sub-goal.

By formalizing the Prime Number Theorem asymptotics ($\pi(x) \sim \mathrm{Li}(x)$) and linking it directly to explicit zero-free regions [53], the AI can systematically and computationally expand the zero-free domain deeper into the critical strip.[54] To avoid "reinventing the wheel," the agent can access decades of pre-formalized lemmas on Fourier analysis, complex contour integration, and Jacobi's theta functions already residing natively in Mathlib.[50] The agent generates mathematical proofs by navigating a massive, discrete tree of logical deductions, bounded entirely by the compiler's strict type-checking.

Phase 4: Operational Directives for the Agent

To operationalize the theoretical paradigms discussed above, the autonomous AI system is issued five highly specific, rigorously prioritized mathematical directives. Each directive outlines a distinct "angle of attack," leveraging different computational strengths—from pattern recognition and high-dimensional gradient descent to formal tactic generation and combinatorial sieving.

Directive 1: Formal Refutation via Primorial Optimization (Nicolas Criterion)

Objective: Disprove the Riemann Hypothesis by computationally discovering a highly composite integer or primorial that violates established arithmetic equivalences. **Methodology:** The agent will execute a distributed, highly parallelized search targeting the Nicolas Criterion: $\frac{N_k}{\varphi(N_k)} > e^{\gamma} \ln(\ln N_k)$. Theoretical constraints explicitly suggest that any potential counterexample must exist at a primorial $N_k = \prod_{i=1}^{k} p_i$ where the k -th prime $p_k > 10^8$.[13] The agent will utilize heavily optimized prime sieve algorithms, coupled with arbitrary-precision integer representations, to compute absolute bounds on the Chebyshev function $\theta(p_k)$ relative to Mertens' theorems. By operating strictly on integers, the agent avoids floating-point inaccuracies associated with complex integration. **Sign of Success:** The algorithm outputs a validated prime p_k and the corresponding totient calculations such that $N_k/\varphi(N_k) \leq e^{\gamma} \ln(\ln N_k)$.[13] Concurrently, the agent must generate a Lean 4 formal proof script certifying the arithmetic computation, providing an immediate, undeniably verified counterexample to RH.

Directive 2: Trace Operator Anomaly Detection on the Semilocal Sonin Space (NCG)

Objective: Isolate a counterexample to Weil positivity utilizing Alain Connes' Noncommutative Geometry scaling Hamiltonian. **Methodology:** The AI will engage with the deep structures of algebraic geometry over characteristic one. By operating on the semi-local adele class space, the agent will computationally analyze the trace formula associated with the scaling Hamiltonian.[27] The agent is mathematically tasked with generating exotic, highly oscillatory test functions $g(x)$ within the semi-local Sonin space. These are specifically functions $g \in C_c^{\infty}(\mathbb{R}_+^{\times})$ whose Fourier transforms precisely vanish at $\pm i/2$.[32] The algorithm will computationally integrate these generated test functions against the local distributions W_v . **Sign of Success:** The agent successfully identifies a specific, smooth test function with compact support within the Sonin space that results in a strictly positive evaluation of the right-hand side of the Riemann-Weil explicit formula: $\sum_v W_v(g * g^{\sharp}) > 0$.[27] Finding a test function that demonstrably breaks Weil positivity immediately and geometrically falsifies the Riemann Hypothesis.

Directive 3: L^2 -Optimization of Vasyunin Sums (Beurling-Nyman-Báez-Duarte)

Objective: Construct a sequence of Dirichlet polynomials that radically accelerates the distance decay in the Beurling-Nyman functional approximation, aiming for a formalized constructive proof of RH. **Methodology:** The agent will model the Báez-Duarte formulation as an infinite-dimensional deep reinforcement learning task. The objective is to approximate the indicator function $\chi_{(0,1)}$ within the continuous Hilbert space $L^2(0, \infty; t^{-2}dt)$ using linear combinations of the fractional part functions $\sum a_k\{t/k\}$.[35] The agent will construct minimizing Dirichlet polynomials $A_N(s)$, systematically training a neural network to predict the optimal Vasyunin sum coefficients a_k that minimize the distance d_N^2 . **Sign of Success:** The AI discovers a deterministic or recursive algorithm to generate coefficients a_k such that the squared distance d_N^2 decays at a rate strictly faster than the currently established $\mathcal{O}(1/\log N)$ or $(\log\log N)^{5/2+o(1)}\,(\log N)^{-1/2}$.[38] If the agent can automatically synthesize a proof script in Lean 4 demonstrating that the limit $\lim_{N\to\infty} d_N^2 = 0$ holds universally using this novel sequence, it will have successfully proven the Riemann Hypothesis via functional density.

Directive 4: Adversarial Universal Approximation in the Critical Strip (Voronin)

Objective: Exploit Voronin's Universality Theorem to formally map the zero-free regions of the critical strip and mathematically separate the chaotic limits of $\zeta(s)$ from its rigid symmetries. **Methodology:** The agent will construct a specialized mathematical Generative Adversarial Network (GAN). The Generator network will design highly erratic, non-vanishing continuous functions $f(s)$ defined strictly on compact subsets K inside the right half of the critical strip ($1/2 < \mathrm{Re}(s) < 1$). The Discriminator network will concurrently attempt to approximate $f(s)$ using purely imaginary vertical translations $\zeta(s + i\tau)$ as dictated by Voronin's theorem.[42] The system will systematically attempt to "trick" the zeta function into approximating a function that inherently pushes a root infinitesimally close to the compact subset boundary. **Sign of Success:** The agent mathematically identifies a rigorous structural barrier separating the universal approximations from actual zero locations. Success is denoted by the extraction of a formalized Lean 4 sub-lemma proving that no universal approximation sequence $\zeta(s+i\tau)$ can force an actual root to remain inside the target domain K in the limit $\tau \to \infty$. This would serve as a critical, verifiable lemma in progressively widening the proven zero-free region toward the critical line, contributing sequentially to a definitive proof.

Directive 5: Gradient Ascent on the Electrostatic Potential of the Hardy Z-Function

Objective: Identify a zero located strictly off the critical line by mapping the zeros to a Dyson Coulomb gas and computationally maximizing electrostatic repulsion. **Methodology:** The agent will treat the zeros of the Hardy Z-function $Z(t)$ as a high-fidelity physical particle simulation. Focusing strictly on the specific topological locations known as "bad Gram points" (where Gram's Law mathematically fails [47]), the agent will train a deep neural network to evaluate the highly non-linear discriminant function $\Delta_n(a)$ up to arbitrary precision. Using gradient ascent, the agent will perturb the simulated zero positions to maximize the second-order Dyson-like repulsion term between consecutive zeros, testing the elasticity of the critical line. **Sign of Success:** The agent forces a mathematical electrostatic "rupture." This manifests as identifying a specific integer n for a bad Gram point g_n where the higher-order terms of the discriminant yield a violation of the corrected law: $(-1)^n \Delta_n(1) \leq 0$.[47] The convergence of the loss function corresponding to a zero trajectory escaping the $\mathrm{Re}(s) = 1/2$ boundary will indicate a high-probability region. The agent will then execute a targeted, ultra-high-precision computational verification using the Riemann-Siegel formula to confirm the existence of a zero off the critical line.

By executing these five operational directives in parallel, an autonomous AI system transcends isolated heuristic searches. It fundamentally integrates the absolute deductive rigor of formal verification architectures with the unbounded, high-dimensional pattern-recognition capabilities of deep reinforcement learning, mapping pure mathematical abstraction onto quantifiable, programmable state spaces. Whether culminating in the rigorous closure of the critical line or the discovery of a profound spectral anomaly, this operational framework guarantees a strategically optimal exploration of the Riemann Hypothesis.

Karya yang dikutip

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